

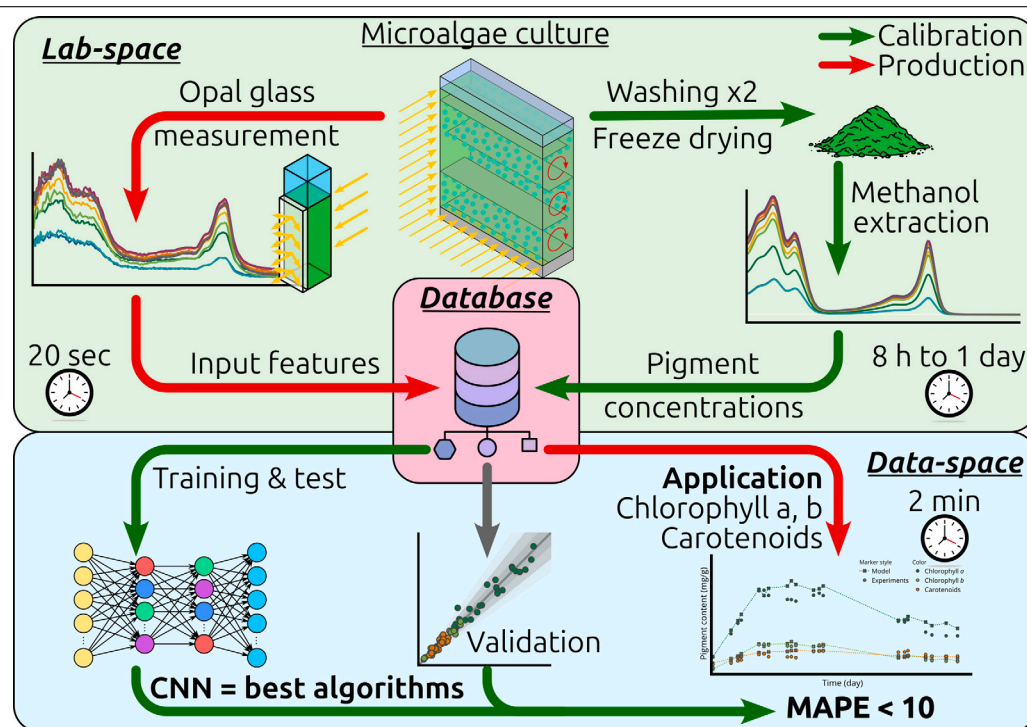


# Opal glass method, an extraction-free way to access microalgae pigment content

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## GRAPHICAL ABSTRACT



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## ABSTRACT

A new extraction-free and non-destructive method to evaluate pigment content (chlorophyll a, chlorophyll b, and total carotenoids) in microalgae cells is introduced. It relies on the use of a spectrophotometer and an optical diffuser, together coined as the opal glass method. First, the practical aspects of the technique are consolidated (ease of setting up, usage of disposable cuvettes, speed of data acquisition, ...). Then, a database is created to link opal glass spectra to microalgae pigment contents obtained by a methanol-extraction method. Partial Least Squares (PLS - machine learning) and Convolutional Neural Networks (CNN - deep

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Extraction-free  
Machine learning

learning) are investigated to retrieve cell pigment content from opal spectra. Data preprocessing (filtration, augmentation/derivative-based feature engineering, signal scaling) is investigated in a systematic manner for both algorithms. In addition, two CNN architectures are tested, and hyperparameters are optimized for each possible combination. Overall, PLS shows merit, but sometimes offers some misprediction, resulting in a 10 to 15 Mean Absolute Percent Error (MAPE). On the contrary, the CNN achieves a stable sub-10 MAPE, making it the tool of choice to link opal glass spectra and cell pigment content. Finally, a learning curve analysis is carried out to demonstrate that the number of samples required to train the algorithms remains manageable (60–90 for the PLS, 80–150 for the CNN). Overall, this technique reduces pigment quantification from several hours (at best) to less than two minutes and opens the way to online reliable microalgal pigment content monitoring.

## 1. Introduction

Cell pigment content plays an important role in monitoring microalgal cultures. On the one hand, they provide valuable information on cellular physiological status, especially related to the cell nitrogen status. Indeed, nitrogen deprivation promotes chlorophyll diminution through reallocation mechanisms, whereas nitrogen replenishment restores chlorophyll biosynthesis and can induce cell regreening (Grimme and Porra, 1974; Pyliotis et al., 1975). On the other hand, pigment composition directly contributes to the commercial value of the resulting biomass, with higher pigment contents generally translating into greater market value (Barros et al., 2019). This economic interest is particularly pronounced for carotenoids, among which lutein stands out for its health benefits and potential for use in microalgae as a dietary vector (Camarena-Bernard and Pozzobon, 2024).

Yet, no matter the intent of the process (from scientific study to pigment production), a fast, reliable, and inexpensive pigment quantification method is desirable. To date, no method is able to achieve these three desirable traits at once. Trade-offs always have to be resorted to. Out of honesty, and unveiling the conclusion of this article, the proposed method also features trade-offs. Among the available methods, two major ones are spectrophotometric methods and HPLC methods.

The spectrophotometric methods have been honed since the end of the twentieth century by the scientific community. Here, the works of Wellburn, Lichtenthaler, and Porra are to be acknowledged (Wellburn, 1994; Porra, 1990; Lichtenthaler and Wellburn, 1983; Lichtenthaler and Buschmann, 2001). Yet, while spectrophotometric methods are efficient, low-cost, and straightforward to implement, they face three limitations. Firstly, the conventional version of these methods does not allow for the discrimination of individual carotenoid species, reporting them as the sum of xanthophylls and carotenoids (a.k.a. “x+c” or total carotenoids). This comment is to be modulated by recent development allowing to resolve lutein, violaxanthin, and zeaxanthin (Bayomie and Pozzobon, 2026; Pozzobon and Camarena-Bernard, 2022). Yet, this improvement is still far from what an HPLC system can achieve. Secondly, most of the time, the samples have to be washed and freeze-dried. Indeed, while faster methods exist, freeze-drying was shown to be the method that preserves cell pigments the most, while achieving similar moisture removal (below 1%w) (Aljabri et al., 2023). While far from being challenging, this step takes time (8 h at best), preventing immediate feedback upon sample withdrawal from the culture system. Thirdly, the extraction of the pigments involves potentially toxic chemicals (e.g., methanol), for which the minimization of use is always welcome.

The HPLC methods are considered as references as they resolve the compounds on a chromatographic column before quantifying them individually. However, these very enticing traits come with several drawbacks. Firstly, it is longer than a spectrophotometric method. Indeed, in addition to the aforementioned steps, a classical analysis lasts between 30 min (assuming the apparatus is operational and available) and several hours if a sequence is in progress. Secondly, an HPLC system is costly and requires expertise to design the procedure (solvent, columns, gradients, ...), operate, and maintain the system. Thirdly, as with spectrophotometric methods, their operation involves potentially toxic chemicals.

In this landscape, the question of why scholars and engineers resort to these methods while the microalgae's color is somehow so evident, may arise. It can be explained by the intertwining of a physical phenomenon (light scattering) and biological considerations. Indeed, from a physical point of view, a microalgae culture is a suspension of small particles of which the diameters allow them to interact in a complex way with light (Hulst, 1957). Consequently, direct pigment content measurement is tied to cell size, the ratio of the indices of refraction between the cells and the medium, the cell light scattering pattern (a.k.a. phase function or Green function) (Lehmuskero et al., 2018), measurement device internal geometry, but also to pigment biological arrangement, such as their packing and association with photosynthesis-related proteins.

Despite the underlined complexity of the task, some authors managed to drive the technique forward. Among them, Ritchie and Sma-Air proposed an elegant method capturing the light scattered by a sample using an integrating sphere (Ritchie and Sma-Air, 2020). Yet, the cost of the apparatus and the need to set it up every time (assuming a shared spectrophotometer) limit its practical applicability. Another avenue was explored by Myers et al. which consists of limiting the refraction indices ratio between the cells and the medium using Bovine Serum Albumin (a.k.a. BSA) (Myers et al., 2013). A positive aspect of this technique is that it adapts directly to an existing spectrophotometric system while necessitating, *a priori*, very limited extra work. The last option worth mentioning is the one introduced by Shibata et al. in the 50's: increasing the system internal scattering so that the microalgae one is rendered negligible (introduced in Shibata et al., 1954, and detailed very pedagogically in Shibata, 1959, extended to plant leaves in Shibata, 1973). From a practical perspective, it consists of placing an optical diffuser in the spectrophotometer beam (e.g., oil-dripped paper), so that the additional scattering capability deviates off-collimated light from the detector (with and without microalgae suspension). Consequently, at the cost of the loss in terms of light impinging on the detector, the distortion of the spectral baseline (originating from scattering) is nullified. Despite its ease of use, very few articles mention the technique. For example, Masojidek et al. used it (with paper) to acquire cell pigment cross-section over the course of a culture (Masojidek et al., 2021). But the most advanced example might be the work of Hervey et al. who used a very specific dual chamber cuvette to reproduce the setup (Hervey et al., 2022). They demonstrated that the system could reach parity with an integrating sphere setup. Still, despite its elegance, it also poses several challenges. The second compartment of the cuvette is to be filled with controlled suspended titanium oxide particles, potentially rendering the system vulnerable to aging, but also requiring washing and reusing the cuvette from sample to sample, hindering the practical applicability.

In this context, this work aims to bridge all these elements together to deliver an easy-to-use, extraction-free, non-destructive, on-the-spot microalgae pigment measurement technique. To do so, it will make use of Shibata's technique (referred to as opal glass technique) and demonstrate that the obtained spectra can be linked with a reasonable level of confidence ( $\pm 10\%$ ) to the results obtained by pigment extraction and spectrophotometric quantification. Data processing will be handled by both linear and non-linear machine/deep learning. The first is extremely fast, while the second is more computational power

demanding but allows for achieving higher performance. Finally, a learning curve analysis will be carried out to inform the readers on the number of samples required to train the technique and assess if it is reasonable to deploy in their laboratory (Viering and Loog, 2023).

## 2. Data generation & Acquisition

### 2.1. Cell cultivation

Data were produced in the context of trophic conversion experiments similar to those presented elsewhere (Levasseur and Pozzobon, 2025). As these experiments underline the sizable phenotypic plasticity of the cells while transitioning from a heterotrophic - low-pigment - phenotype to a phototrophic - high-pigment - one, they are particularly well-suited to create a database featuring maximum contrast between the data points.

In more detail, *Chlorella vulgaris* 211-11b, bought from SAG Culture Collection, Germany, and *Chlorella sorokiniana*, kindly shared by Biorea, France, were used for this study. They were cultivated in Bold's Basal Medium (Andersen and America, 2005) with tripled nitrogen content (classically referred to as B3N), with 20 g/L of glucose before being switched to a regular, glucose-free, B3N medium for the trophic conversion experiments. The strains were used in two different sets of investigations. For *Chlorella vulgaris*, the runs were carried out in flasks (250 mL, 50 mL working volume, 100 rpm, 25 °C, 45  $\mu\text{molPhotonPAR}/\text{m}^2/\text{s}$ ). For *Chlorella sorokiniana*, the runs were carried out in a flat panel photobioreactor (10 L, 1 VVM, 25 °C), under incident illuminations between 45 and 150  $\mu\text{molPhotonPAR}/\text{m}^2/\text{s}$ .

### 2.2. Pigment quantification

Samples were withdrawn at various time points during the conversion trophic process (thrice a day, most of the time). Among them, one was dedicated to pigment extraction and profiling, while the other was used for immediate opal glass analysis (see next section). In the case of pigment profiling, samples were washed twice with Milli-Q water by centrifugation (11 000 rpm, 10 min, 4 °C) within minutes after withdrawal. The pellet was then frozen (−20 °C) and freeze-dried for 24 h under primary desiccation using a freeze dryer (Christ Alpha 12 LD+). Owing to the quantity of biomass obtained through the initial heterotrophic cultivation stage, each pellet contained a minimum of 2.07 mg of dry microalgal matter and a maximum of 4.36 mg, ensuring enough biomass was available to achieve a reliable pigment quantification. For extraction, the pellet was homogenized in 5 mL of pure methanol using an MP Biomedicals FastPrep-42 bead beater. The mixture was then incubated in a water bath at 60 °C for 20 min to ensure complete pigment extraction, considering the known recalcitrance of the *Chlorella* genus (Porra, 1990). The supernatant was filtered through 0.22  $\mu\text{m}$  filters before spectrophotometric analysis (Shimadzu UV-1800 UV/Vis, dual beam) in quartz cuvettes. Throughout the entire protocol, samples were protected from light to prevent pigment degradation.

Full absorbance spectra from 340 to 800 nm were recorded with a resolution of 0.2 nm, and pigment concentrations (mg/L) were calculated based on Welburn's high-resolution equations (Wellburn, 1994) with Lichtenthaler's processing guidelines (Lichtenthaler and Buschmann, 2001):

$$[\text{Chlorophyll } a] = 16.72A_{665.2 \text{ nm}} - 9.16A_{652.4 \text{ nm}} \quad (1)$$

$$[\text{Chlorophyll } b] = 34.09A_{652.4 \text{ nm}} - 15.28A_{665.2 \text{ nm}} \quad (2)$$

$$[\text{Total carotenoids}] = (1000A_{470 \text{ nm}} - 1.63[\text{Chlorophyll } a] - 104.96[\text{Chlorophyll } b])/221 \quad (3)$$

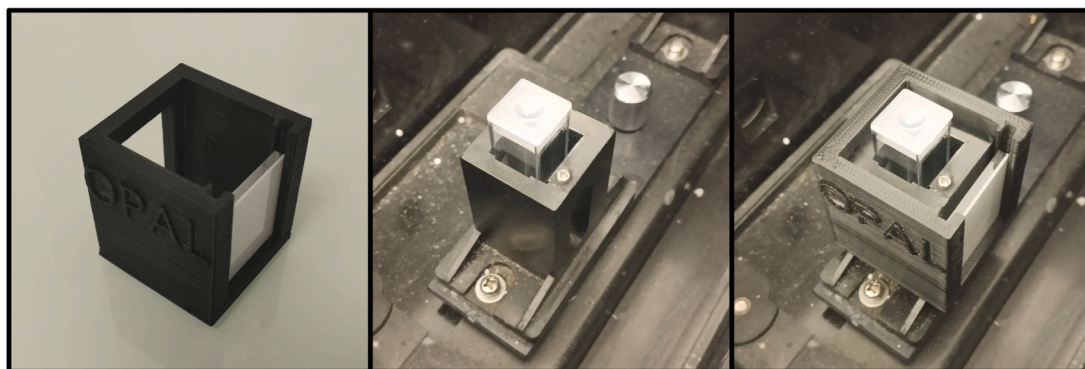
### 2.3. Opal spectra acquisition

Every time a sample was withdrawn from a culture, the cell suspension was analyzed using the opal glass method concomitantly with the washing of the paired sample. The setup was made of a removable 3D-printed holder (Fig. 1 - Left) supporting the opal glass material (Edmund optics - Opal glass 20 × 20 mm<sup>2</sup>). The holder could be effortlessly, yet securely, tied to the conventional spectrophotometric setup (Fig. 1 - Middle & Right), allowing for modulating the signal outgoing from the cell suspension. While numerous calibration details are presented in the Supplementary Materials, only the most important ones will be summarized here. Cell suspensions were placed in 3 mL plastic cuvettes (1 mL is not advised). Spectra were acquired (Shimadzu UV-1800 UV/Vis, dual beam) from 340 to 800 nm, with a 2-nm resolution. Only the 400–800 nm range was used, as the blue end of the spectrum was too noisy (also observed by Hervey et al., 2022). When the maximum absorbance was above 0.5, the samples must be diluted to fall back below this threshold (see Supplementary Materials for linearity guideline elaboration). While dilution limits were not investigated, the minimal absorbance recorded at 800 nm in this work was 0.076, which did not induce any acquisition problem. After acquisition, the signals were numerically processed using the Savitzky-Golay algorithm for noise removal and derivation (orders 1 and 2) purposes (see Supplementary Materials) (Savitzky and Golay, 1964). Finally, the kind reader may note that, once analyzed, the sample can be used for other analyses such as flow cytometry or fluorometry.

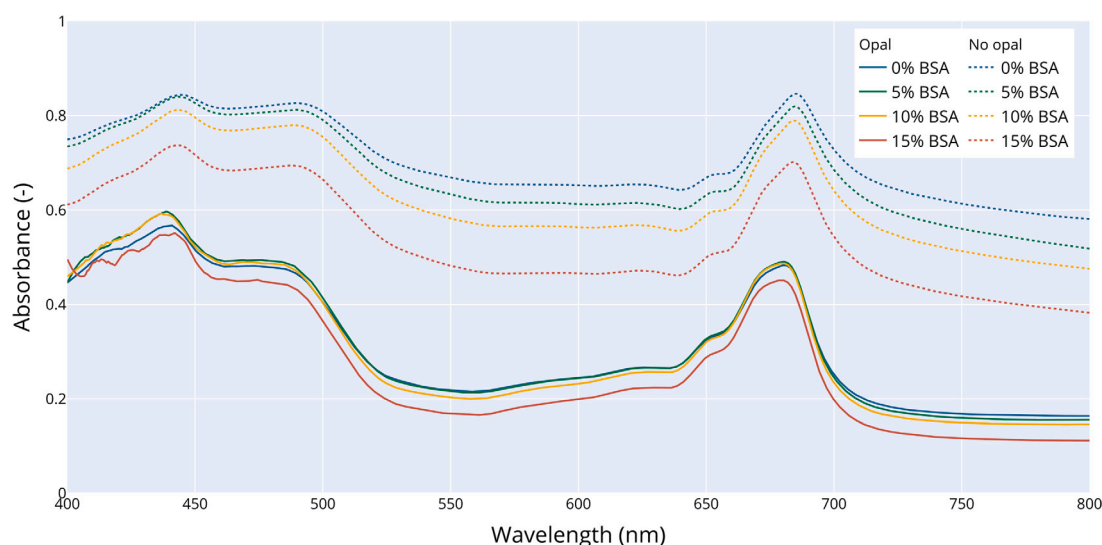
### 2.4. Comparison with BSA method

Going further, it is important to compare the results obtained by the opal glass method and the addition of BSA to the measurement medium, as they intend to deliver the same benefit: managing light scattering. To do so, a systematic screening was carried out on a given sample. The cell suspension was loaded with a specific amount of BSA (w/v, in the measured suspension, milliQ water was used to ensure a constant dilution factor). The obtained mixtures of microalgae suspension, BSA, and milliQ water had a BSA loading ranging from 0 to 15%w/v. They were measured with and without the opal glass to also assess for a possible synergy. Before analyzing the results, it is important to point out some technical difficulties when handling the BSA solution. Indeed, samples containing BSA are extremely prone to foaming and demand gentle homogenization, making them time-consuming to prepare. In addition, the time stability of the BSA mother solution was not investigated and remains a topic of question. Therefore, if there were any benefits to using BSA in conjunction with (or instead of) the opal glass technique, it would come at a practical cost. Finally, BSA alone induces a baseline shift to be compensated for. While this would not represent an extra step in the process (as a baseline against water is performed), one has to be aware of the necessity.

Fig. 2 reports the results of the screening. First of all, for each and every configuration, a dedicated baseline was performed and subtracted. This step enhanced the quality of the results (the interested reader is referred to the Supplementary Materials for the uncorrected version). Second, the BSA-only measurement exhibits a higher span in terms of absorbance compared to the one featuring the opal glass method. This results in a lower signal-to-background ratio. Taking the height of the chlorophyll peak as a proxy for signal and the average over the cell wall region (730 to 800 nm) as a proxy for background, the BSA loading achieves a  $1.55 \pm 0.18$  ratio, over all the conditions, while the opal glass method achieves  $3.29 \pm 0.38$ . Acquired spectra also exhibit a higher sensitivity to the BSA-loading, which is virtually nonexistent when the opal glass is used (for BSA loading of 10% or less). One last negative aspect of the BSA-loading method compared to the opal glass technique is its inability to produce a flat signal in the 730–800 nm range, where pigments are absent. From a quantitative point of view, the slope of the signal is of  $-0.054$  to  $-0.064$  over



**Fig. 1.** Opal glass measurement setup. Left - opal glass and 3D-printed holder. Middle - regular spectrophotometer cuvette setup. Right - opal glass spectrophotometer setup.



**Fig. 2.** BSA baseline corrected signals. Opal signals are filtered with Savitzky-Golay algorithm (order 2, 7-point fitting window).

the 70-nm span for the BSA loading method, versus  $-0.010$  to  $-0.011$  for the opal method, a five-fold reduction accompanied by a dramatic compression of the spread. Therefore, the opal glass method can be deemed superior to the BSA-loading method both in terms of ease of deployment and results quality. Finally, there seems to be no potential benefit in coupling the two methods, as the opal glass signal with 0 to 10% BSA loading was identical.

## 2.5. Resulting database

Once the experiments and manual data curation have been completed, the dataset contains 203 pairs of spectra (opal glass technique and methanol pigment extracts). Out of them, one run (20 samples) was withdrawn to later simulate the field experiment monitoring. Therefore, 183 data points remained for the algorithm training and validation. They were split randomly (80/20) between the two sets. The validation test was kept hidden from the algorithms until the moment to assess their performance. Furthermore, for all the optimization pipelines described below, the training set was divided into actual training and test subsets, using 5-fold cross-validation in order to obtain models as robust as possible.

Once model optimal architectures have been established (using 183 data points for training and validation, plus 20 never-before-encountered data points constituting the test run), in an effort to assess the number of data points required to achieve a given level of performance, a learning curve analysis will be carried out. For this

specific aspect of the manuscript, a bootstrap procedure over the whole dataset (203 data points) was opted for (80% training, 20% validation). [Fig. 3](#) proposes a graphical summary of the data handling protocol.

In addition to pure machine learning consideration, it is important to note that the lowest pigment content value over the whole dataset was 1.34 mg/g. Therefore, it ensures that all the reference values are strictly above 0, which renders the use of most of the conventional metrics possible. Consequently, Mean Absolute Percent Error (MAPE) was used in this work as it is particularly well suited to get a grasp at the performance of a given model (*i.e.*, percentage of deviation), so long as outliers and zero-values are absent, which is the case. Therefore, MAPE was the metric driving the optimization and the reporting of the results. Still, it was complemented by other metrics such as the Root Mean Square Error (RMSE), as it conveys an absolute sense of the deviation (in mg/g). Also, moving from assessing the deviation to assessing the quality of the prediction, the Nash–Sutcliffe Efficiency (NSE) was used as it allows for weighting for outliers owing to its quadratic nature and offers an easy interpretation: close to 1 is good, 0 means using the average value of the observation would have delivered the same prediction, and a negative value is worse than using the average value ([Nash and Sutcliffe, 1970](#)). Finally, the last metric employed was the Concordance Correlation Coefficient (CCC), as it is an advanced metric that quantifies how similar two methods are (*e.g.*, two analyzers processing the same samples, or, in this case, experimental readings and numerical predictions) ([Lin, 1989](#)). A value close to 1 indicates good agreement, values close to 0 or negative values indicate poor concordance due to either systematic bias, reduced correlation, or both.

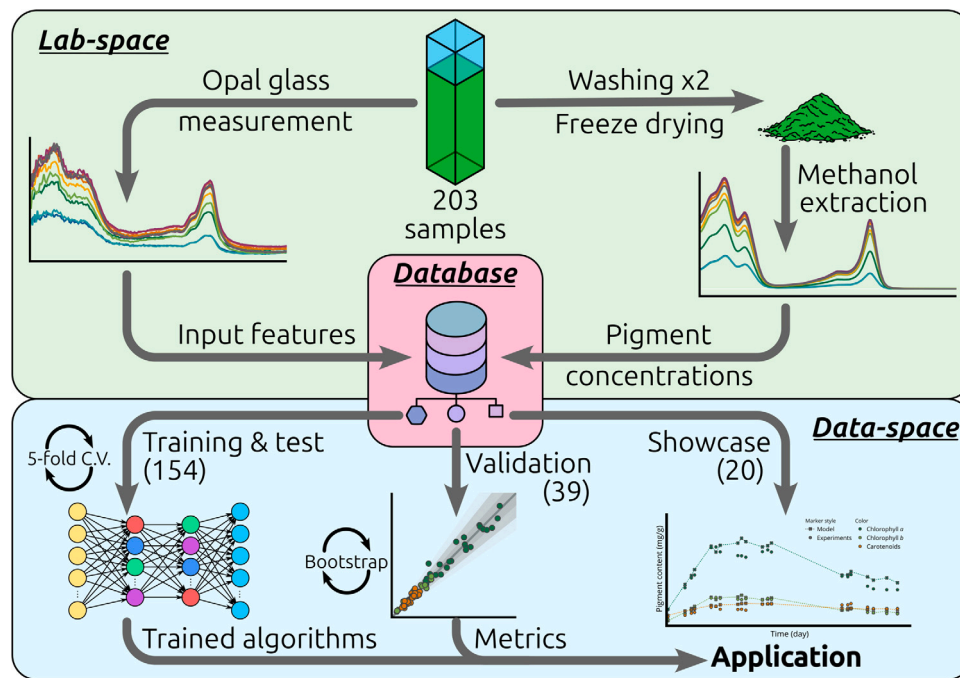


Fig. 3. Graphical summary of the workflow.

### 3. Processing pipeline

First of all, in order to data augment/engineer the data, the Savitzky-Golay was applied to lead first- and second-order derivation of the opal glass spectra (see Supplementary Materials) (Savitzky and Golay, 1964). Indeed, spectrophotometric spectra derivation has long been established as a reference procedure to enhance predictions quality (Ni and Gong, 1997; Cerdà et al., 2022). This procedure increases the size of the input vector from 201 (raw spectra) to 603 features (+402 from derivatives). Then, several preprocessing steps were tested on the first part of the input vector (raw signal) with the hope of minimizing noise while maximizing signal quality. Yet, the relevance of the preprocessing techniques is case- and data-quality-dependent (e.g., the presence of a baseline deviation), and therefore needs to be confirmed through a systematic screening.

#### 3.1. Signal preprocessing

##### 3.1.1. Raw data vs. filtered data

As the opal glass spectra feature a high level of noise in the blue end of the visible spectrum, the Savitzky-Golay algorithm was used to filter the raw signal (see Supplementary Materials). Still, in order to investigate the influence of this step, raw spectra were also included. Consequently, the three following preprocessing steps were performed on both filtered and raw data. All in all it yielded  $4 \times 2 = 8$  data preprocessing combinations: (None, Multiplicative Scattering Correction, Z-scaling, and Pigment-scaling)  $\times$  (Raw and Filtered data).

##### 3.1.2. Multiplicative scattering correction

Multiplicative Scattering Correction (MSC) is an established preprocessing method to correct additive and multiplicative effects within a set of spectra (Geladi et al., 1985). In a nutshell, the method adjusts each spectrum to a reference spectrum (i.e., the average spectrum). Each spectrum ( $x$  vector) undergoes a series of transformations to remove unwanted physical variations (i.e., baseline deviation). The process begins with mean centering:

$$x_i^{\text{centered}} = x_i - \frac{1}{n} \sum_{j=1}^n x_{ij} \quad (4)$$

With  $i$ , from 0 to  $m$ , being the index of the spectra, while  $j$ , from 0 to  $n$ , is the index of the wavelength. The reference spectrum is then calculated as:

$$x_i^{\text{ref}} = \frac{1}{m} \sum_{i=1}^m x_i^{\text{centered}} \quad (5)$$

Then, for each centered spectrum, a linear regression against this reference is performed:

$$x_i^{\text{centered}} = b_i \cdot x_{\text{ref}} + a_i + e_i \quad (6)$$

where  $b_i$  is the multiplicative coefficient related to scattering effects,  $a_i$  is the additive coefficient related to baseline shifts, and  $e_i$  contains the residual information.

The final MSC-corrected spectrum is obtained by:

$$x_i^{\text{MSC}} = \frac{x_i^{\text{centered}} - a_i}{b_i} \quad (7)$$

Which is a spectrum free of the estimated physical interferences.

##### 3.1.3. Centered-reduced/Z-scaling

Standardization or Z-score normalization (referred to as Z-scaling in this article) represents a data normalization preprocessing technique. It is more frequently encountered in the data processing community than in the spectroscopy one, but is advised by some authors in the context of CNN use (Bjerrum et al., 2017):

$$x_{i,j,Z\text{-scaled}} = \frac{x_{i,j} - \mu_j}{2\sigma_j} \quad (8)$$

where  $\mu_j$  is the mean value for a given wavelength over the whole training dataset (183 data points) and  $\sigma_j$  is the associated standard deviation. Standardization enables all variables to operate in the same range by removing scale differences between wavelengths. Therefore, the high absorption and low absorption regions of the spectra show similar magnitudes after this preprocessing. The method demonstrates its best performance when absorbance signals at different wavelengths exhibit substantial variation, which is typical in pigment absorption spectra.

### 3.1.4. Pigment-scaling

The last scaling employed (coined as “pigment-scaling”) comes from the intuition offered by the shape of an opal glass spectrum (Fig. 2). The red end of the spectrum (flat section above 750 nm) could represent the contribution of the cell walls, while the peaks are tied to pigment contents. Therefore, dividing the contribution of the wavelength over the baseline by the absorbance value measured above 750 nm, one would get a preprocessed spectrum already containing a sense of the targeted “pigment per cell” outcome.

$$x_{i,j,\text{Pigment-scaled}} = \frac{x_{i,j} - v_i}{v_i} \quad (9)$$

where  $v_i$  is the estimation of the baseline originating from the cell wall signal, obtained by averaging the opal glass signal above 750 nm.

## 3.2. Machine learning pipeline

Linking a mixture of pigments' spectra to the individual species concentration is a common problem that has already been tackled using machine learning (see Cerdà et al., 2022 for a review). Yet, specific examples can be used to illustrate how different algorithms perform under increasingly complex setups. In the simplest case, peaks of the different chemical species are clearly isolated. The Inverse Least Squares or Principal Component Regression are suited to recover the individual concentrations without any specific preprocessing (Dinç et al., 2001). Then comes the case where a few species overlap. MultiLinear Regression can still deliver good performance, but the features have to be augmented by a derivation preprocessing (Lababpour and Lee, 2006). Finally comes the case at hand, i.e., numerous species of which the spectra overlap. In this configuration, the Partial Least Squares (a.k.a. PLS) approach holds the upper hand, compared to Principal Component Regression and MultiLinear Regression (Wold et al., 2001), owing to its good management of collinearity (Geladi and Kowalski, 1986). Therefore, PLS will be retained as the machine learning algorithm for this study.

The PLS implementation used in this work was drawn from Scikit-learn (Pedregosa et al., 2011) and was applied to the full feature set (i.e., 603 features per data point), per outcome (hence, 4 times PLS1). The number of components (main hyperparameter of the method) was tuned using the training set, within which cross-validation (5-fold) was used to assess for variability in the results. In the graphs, the results are reported as the mean of the cross-validation process (solid lines) and standard deviations (shadings).

Finally, in order to assess which parts of the feature vector are selected by the PLS algorithm and provide some explainability, a Variable Importance in Projection (a.k.a. VIP) analysis was carried out for chlorophyll a prediction. As the significance threshold can be debated (0.8 to 1), the suggestion of Mehmood et al. to opt for a repeated assessment was adopted (Mehmood et al., 2012). Consequently, the VIP scores were derived for each of the folds of the cross-validation procedure and presented as the average and the resulting standard deviation. The significance thresholds were considered as above 0.8 for a significant feature and above 1.2 for a very significant one.

## 3.3. Deep learning pipeline

Deep learning approaches addressing this type of problem remain relatively scarce in the literature. However, broadening the scope to the infrared spectroscopy literature reveals several relevant examples. Bjerrum et al. employed a Convolutional Neural Network (CNN) to predict pharmaceutical tablet dosages from infrared spectra (Bjerrum et al., 2017). Their architecture consisted of two convolutional layers that progressively extracted local spectral features, followed by dense layers that mapped these features to the target dosage. Similarly, Cui et al. proposed a CNN for predicting wheat flour protein content from infrared spectral measurements (Cui and Fearn, 2018). Their model

relied on a single convolutional layer followed by two dense layers for regression, increasing the CNN's capacity to model complex nonlinear relationships. These two designs were retained for the study, and an illustration of their architectures can be found in the Supplementary Materials.

Even though the architectures of the networks are set, optimizing their hyperparameters is a challenge. To tackle, the Optuna hyperparameter optimizing framework was used (Akiba et al., 2019). The design space covered the number of filters, kernel size, kernel stride in the convolutional layer, layer size in the dense layer, dropout rate, regularization (L2) factor, and learning rate (search space detailed in the Supplementary materials). Tree-Structured Parzen Estimator (a.k.a. TPE) was used as an exploration algorithm combined with the Asynchronous Successive Halving Algorithm (a.k.a. ASHA) pruner, as featured in the Optuna showcase article (Akiba et al., 2019). Each optimization consisted of 100 trials, each hosting a nested 5-fold cross-validation over the training set (best configuration usually identified after about 40 trials). Finally, individual CNNs were trained using the Adam optimizer, with a decreasing learning rate (divided by two) when validation loss stalled for 50 epochs or more. In addition, to prevent overfitting, an early-stopping criterion was employed when the validation loss did not decrease significantly (0.01 MAPE) over 100 epochs.

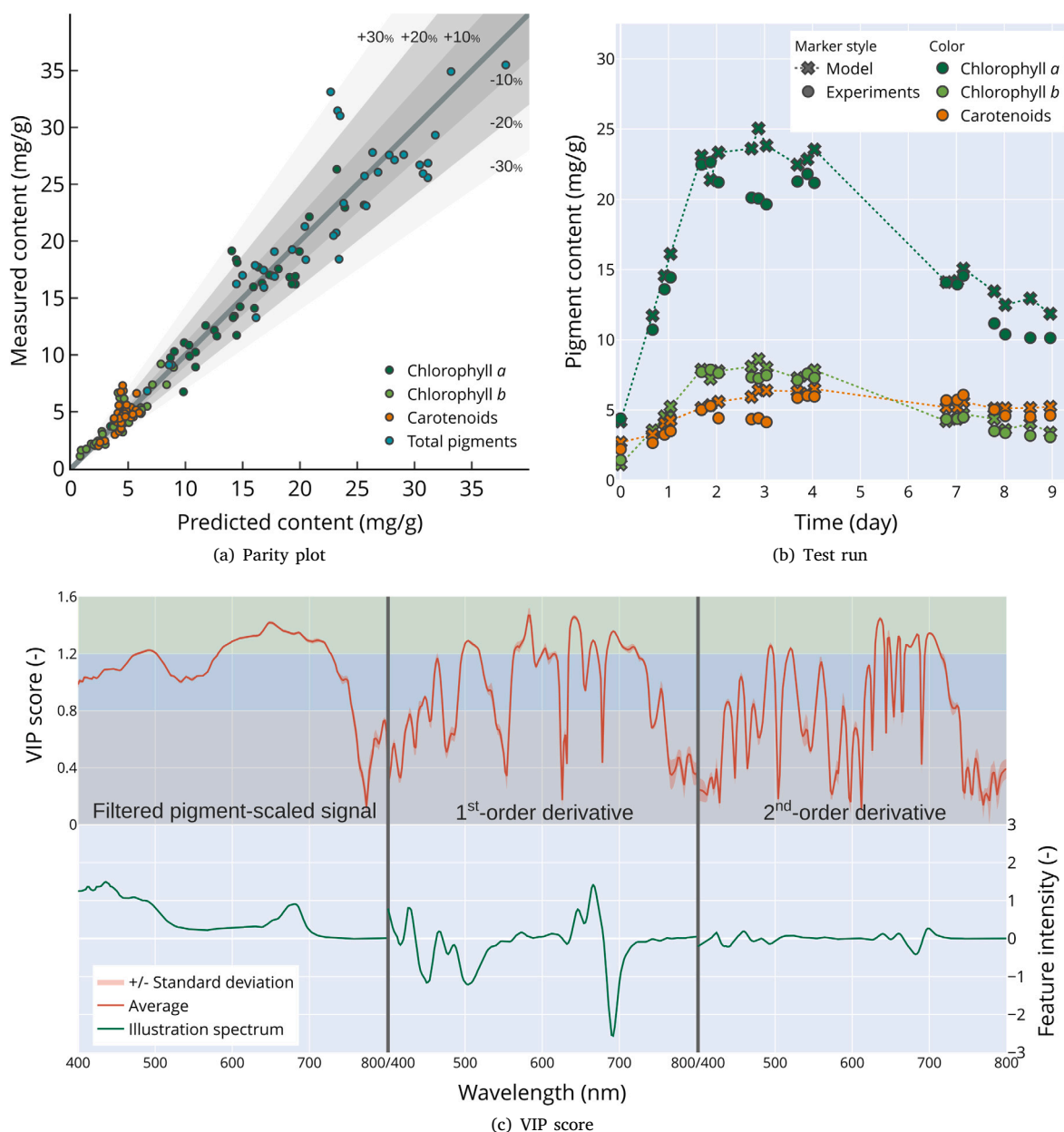
Similar to what was done for the PLS algorithm, an assessment of feature importance was carried out for chlorophyll a prediction. While several methods exist to explore CNNs and get some grasp at their functioning, the majority of them may not possess all the desirable features for such a procedure (Sensitivity and Implementation Invariance) (Sundararajan et al., 2017). To alleviate this problem, Sundararajan et al. introduced the Integrated Gradient approach, which was retained in this work. Furthermore, as underlined by the authors who introduced it, it requires a carefully designed baseline (ideally a lack of signal yielding 0 as prediction), which was easily achieved in this case by using an almost zero feature vector (−0.0202 to be precise).

## 4. Results & Discussion

### 4.1. Partial least squares

The results of the screening aiming at determining the best number of components (preprocessing- and output-wise) are detailed in the Supplementary Materials and briefly summarized here to focus on the optimal pipeline. The cross-validation (5-fold) screening of all the possible preprocessing combinations showed that preprocessing methods effectively impact the results. MSC and Z-scaling exhibit similar trends with high fold-to-fold dispersion. Furthermore, increasing the number of components (from 1 to 10) returns marginal gains (up to 3 components) followed by hindered performance. In contrast, not preprocessing the signal, or using the pigment-scaling procedure, yields curves with a characteristic elbow shape. In these cases, the optimal number of components is 2 for all the outputs. Focusing on quantitative indicators, pigment-scaling delivers marginally better results than the unprocessed signal (−2.44 ± 2.44 MAPE). Also, filtering the signal offers benefit when a high number of components is used (configurations which are not relevant), but does not improve predictions when an adequate number of components is selected (−0.049 ± 0.093 MAPE). Consequently, when PLS is intended to be used, pigment-scaling with filtered signal was used as preprocessing, as this pipeline delivers the lowest MAPE.

Focusing on the best-performing configuration, it achieved 10.69, 15.26, 13.13, and 10.36 MAPE when predicting chlorophyll a, chlorophyll b, total carotenoids, and total pigments, respectively, on the validation set (Table 1). In terms of RMSE, these results correspond to values of 2.05, 0.84, 0.76, and 3.54 mg/g, respectively. Overall, the models showed good predictive performance with NSE of above 0.8, for chlorophyll a, chlorophyll b, and total pigments. The prediction



**Fig. 4.** PLS algorithm performances. (a) - parity plot using data from the validation set. (b) - Performance over one whole biological cultivation. (c) - Chlorophyll a VIP plot using a pigment-scale spectrum as illustration.

of total carotenoids was more difficult, with a NSE of 0.564. This observation is confirmed by the CCC values, above 0.9 for chlorophyll a, chlorophyll b, and total pigments, while being only of 0.703 for the total carotenoids. A sense of the general performance of this workflow is offered by Fig. 4a parity plot. As one can see, most of the points indeed fall into the  $\pm 10\%$  range. Yet, while some points may appear disproportionately underpredicted, the error analysis (see Supplementary Materials) shows that the residuals are overall balanced. Indeed, the highest off-centering (measured as the residual mean-to-residual standard deviation) is 12.0%, for chlorophyll b. Regarding the total carotenoids prediction, one can note that the correspondence between measured and predicted values becomes weak around 0.5 mg/g, confirming visually the limits underlines by the NSE and CCC.

Moving the field testing of the pipeline (Fig. 4b), one can see that the overall trend (rise, plateau, fall, stabilization) is captured for all the pigments. Still, some mispredictions are to be underlined. For example, in the plateau stage (from day 2 to 4), chlorophyll a content

was overpredicted (about +5 mg/g). The same can be observed in the stabilization stage (about +2 mg/g). Total carotenoids predictions also show an appropriate capture of the trend, but with approximate values (too low on the first part of the run, too high afterward). On the contrary, chlorophyll b is relatively well predicted with a deviation below 1 mg/g most of the time. Quantitatively, this leads to MAPE of 11.32, 10.05, and 17.03 for chlorophyll a, chlorophyll b, and total carotenoids, respectively, on the field trial set. While this biological trial can be considered an adversarial case to illustrate the capabilities of the pipeline, it is also an honest one that underlines the limitations of the tool, despite overall good average performance (10 to 15 MAPE).

Detaching from pure performance analysis, Fig. 4c examines the features selected by the PLS algorithm to yield its prediction (in the case of chlorophyll a). As one can see, the algorithm makes heavy usage of the regions of the opal spectrum corresponding to pigments (e.g., the 470 nm region, distinctive peak of carotenoids, the 600–730 nm region, where the two chlorophylls contribute). When it comes to the

first- and second-order derivatives, the PLS algorithm focuses on the central peaks while disregarding the lower and higher ends. Finally, it is interesting to note that the second-order derivative substantially contributes to the final predictions, which will not be the case for the CNNs.

#### 4.2. Convolutional neural network

While the PLS algorithm features an extreme ease of use and robust predictions, it still fails to capture some subtleties, that would have allowed it to closely match the concentrations obtained using a pigment extraction and drive the MAPE below 10. Therefore, the use of algorithms featuring a higher number of degrees of freedom and capable of non-linear relationships was investigated in the form of two CNN architectures (drawn from Cui and Fearn, 2018 and Bjerrum et al., 2017). Yet, in addition to choosing an architecture, operating a CNN requires determining several hyperparameters, per pre-processing of the opal spectral data. This procedure was led using the Optuna hyperparameter optimization framework, and detailed results are available in the Supplementary Materials. For the sake of legibility, only the conclusions will be presented here. The results are presented as MAPE value obtained after a 5-fold cross-validation procedure to ensure robustness of the presented values.

First of all, when considering all the preprocessing combinations, the two architectures performed similarly (8.55 MAPE for Cui's architecture, 8.53 for Bjerrum's one) and reacted similarly to preprocessing. For both architectures, filtered and non-transformed signals yield the best performances, while Z-scaling and MSC, when not filtered, deliver the worst performances. Still, it is important to note that the worst performance remains totally acceptable with a MAPE at 11.43 (Bjerrum's architecture, not-filtered, MSC, chlorophyll b output). Qualitatively, the underperformance of MSC and Z-scaling preprocessings can be explained by the lack of baseline distortion in the opal glass spectra and their relatively low contrast across the visible spectrum (1 to 2.5 ratio)

In order to dive into the results while maintaining a reasonable word count, the focus will now be set on Bjerrum's architecture. In detail, and using an systematic analysis framework, filtering the signal reduced the MAPE by 0.20. Not preprocessing the signal yielded the best performance, followed by pigment scaling (+0.40 MAPE), Z-scaling (+1.33 MAPE), and MSC (+1.53 MAPE). The quality of the prediction was also pigment-dependent, with the best performances achieved for chlorophyll a, then total pigment (+0.08 MAPE), total carotenoids (+1.21 MAPE), and chlorophyll b (+1.92 MAPE). Overall, it can be concluded that both architectures are performing well, that the opal glass signal should be filtered but not preprocessed. Finally, as one has to select one combination to illustrate the result, the best performing one is retained: Bjerrum's architecture combined with data filtration, but no scaling.

Fig. 5a presents the results obtained with the retained deep learning pipeline on the validation sets (Table 1). As one can see on this parity plot, the prediction at low concentrations is improved compared to the PLS algorithm. Still, some slight underpredictions persist for chlorophyll a and total pigment content (driven by chlorophyll a contribution). Moving from qualitative observations to quantitative assessment, the pipeline achieved 8.14, 9.61, 8.74, and 7.69 MAPE when predicting chlorophyll a, chlorophyll b, total carotenoids, and total pigments, respectively, effectively breaching the  $\pm 10\%$  barrier set as an objective for this study. Regarding the other metrics, compared to PLS, a reduction in RMSE is observed ( $-0.08$ ,  $-0.18$ ,  $-0.11$ , and  $-0.33$  mg/g, respectively), while improvements in NSE and CCC are more contrasted, owing to the already very good performance of the PLS algorithm. Indeed, the prediction of chlorophyll a, chlorophyll b, and total pigments was marginally enhanced. On the contrary, cell total carotenoids scores increased by 0.117 and 0.86, NSE and CCC, which

is considerable, even though it does not allow for reaching a CCC of 0.9 or higher.

Take a step back from this success, one should note that it is normal that these values somewhat differ from the previously mentioned ones, as they are not obtained on the same dataset: training dataset with 5-fold cross-validation previously, validation dataset, never presented to the pipeline now. The fact that performance is maintained on data never encountered is a token of the robustness and generalizability of the obtained CNN. Analyzing the errors (see Supplementary Materials) shows that the residuals are less balanced than in the case of the PLS algorithm. Indeed, the CNN achieved narrower error distributions by off-centering them with a maximal residual mean-to-residual standard deviation of 35.6%, for the total pigment content.

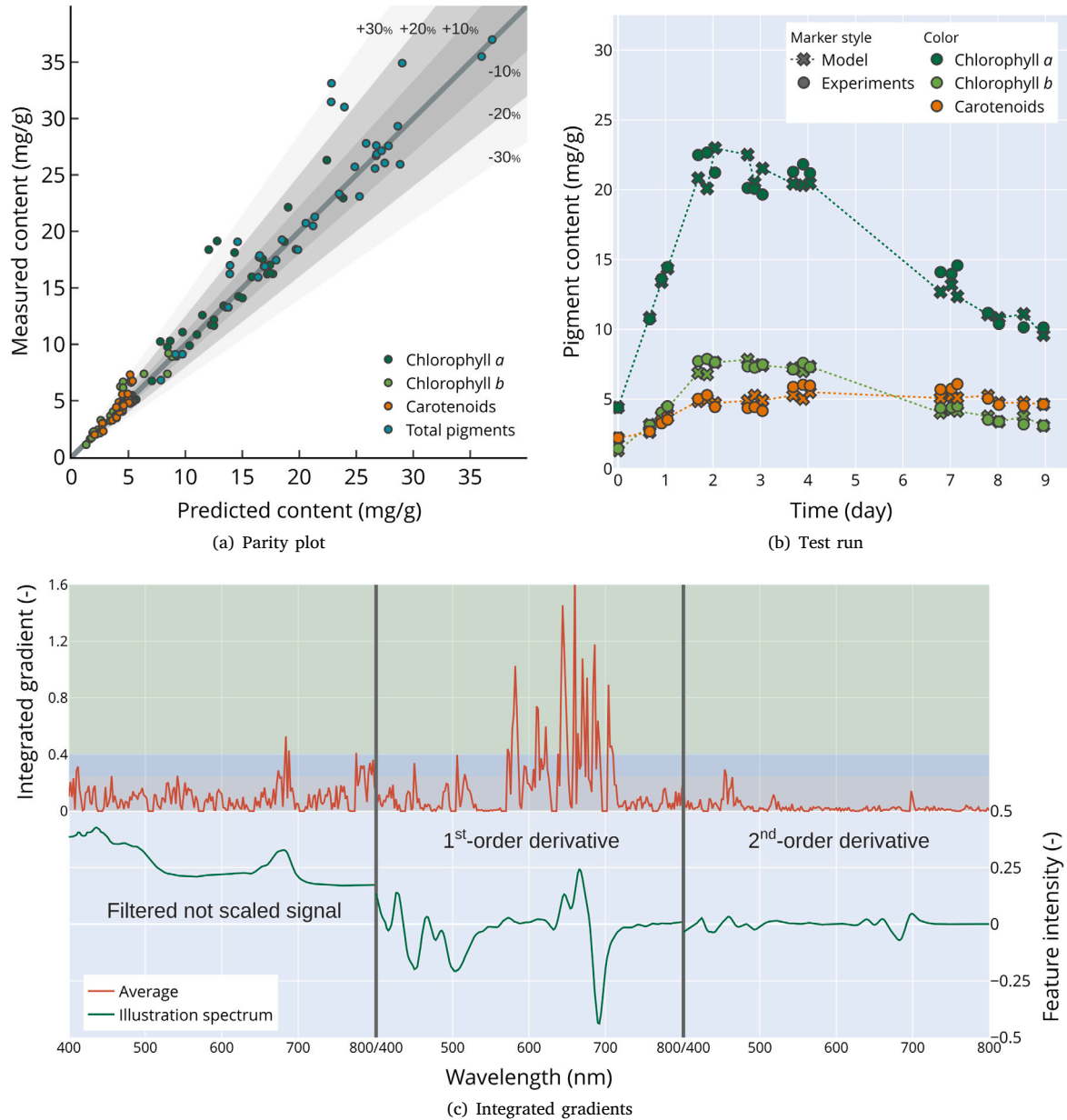
Moving to the field testing of the pipeline (Fig. 5b), resorting to a CNN offered definitely improved prediction over a PLS algorithm. The CNN is able to faithfully predict the three pigment contents over all the phases of the experimental trial and delivers stellar performances (5.93, 5.34, and 7.94 MAPE for chlorophyll a, chlorophyll b, and total carotenoids, respectively).

Finally, as for the PLS algorithm, it is interesting to derive some explainability on how the prediction is formed. Fig. 5c presents the Integrated Gradient obtained with the best performing pipeline (in the case of chlorophyll a). As one can see, the algorithm focused primarily on the first-order derivative (575 to 725 nm), followed by the chlorophylls region of the opal spectrum (650 to 700 nm), the carotenoid region of the first- and second-order derivative, and the cell wall region (750 nm and above). The importance of these regions can be explained as follows: the first four allow for the quantification of the different pigments in the extracts, but as the targeted information is relative to the cell density (*i.e.*, pigment contents in mg/g), the algorithm relies on the fifth region (750 nm and above) to quantify cell density. While this mechanism could be expected to have a sense of generality (across different biological materials), the specific regions will surely vary from one type of sample to another, as they exhibit different pigment profiles and cell wall contributions to the overall signal. One last comment is that the CNN makes almost no use of the second-order derivative, which opens the way to a potential reduction of the feature complexity by removing it from the dataset.

#### 4.3. Learning curves

The final step in this research is to answer the question of how many data points are required to train the selected algorithms so that they achieve the desired level of performance. To tackle this question, the learning curve framework (reviewed by Viering and Loog, 2023) is of help. In a nutshell, the database agglomerated for this work will be subsampled randomly to contain the desired number of data points (from 10 to 200, increments of 10). This subsample is then split into training (80%) and validation (20%) sets. The entirety of the training set is used to fit the algorithm (PLS or CNN), and the validation set is used to assess performance. For each of the tested number of data points, the process is repeated in a bootstrapping process (500 times for each outcome, for both PLS and CNN). Afterward, the distributions of performance are analyzed and fitted with a phenomenological law (decreasing exponential or power law, if well-behaved) to extract the expected asymptotic performance and rate of convergence towards this asymptote.

Fig. 6a presents the learning curves for the PLS algorithm. As one can see, the range of the achievable predictions is broad when data points are limited, but converges monotonically afterward. The asymptotic performances are 10.5, 13.9, 12.3, and 10.4 MAPE for chlorophyll a, chlorophyll b, total carotenoids, and total pigments, respectively. The convergence rate varies from 19.2 to 29.0 data points. As it follows a first-order dynamics, it can be assessed that 60 to 90 data points (at 3 times the decay rate) are required to reach 95% of the



**Fig. 5.** CNN algorithm performances. (a) - parity plot using data from the validation set. (b) - Performance over one whole biological cultivation. (c) - chlorophyll a integrated gradient plot using a filtered, not scaled, spectrum as illustration. Green shading - 95th percentile. Blue shading - 90th percentile.

**Table 1**

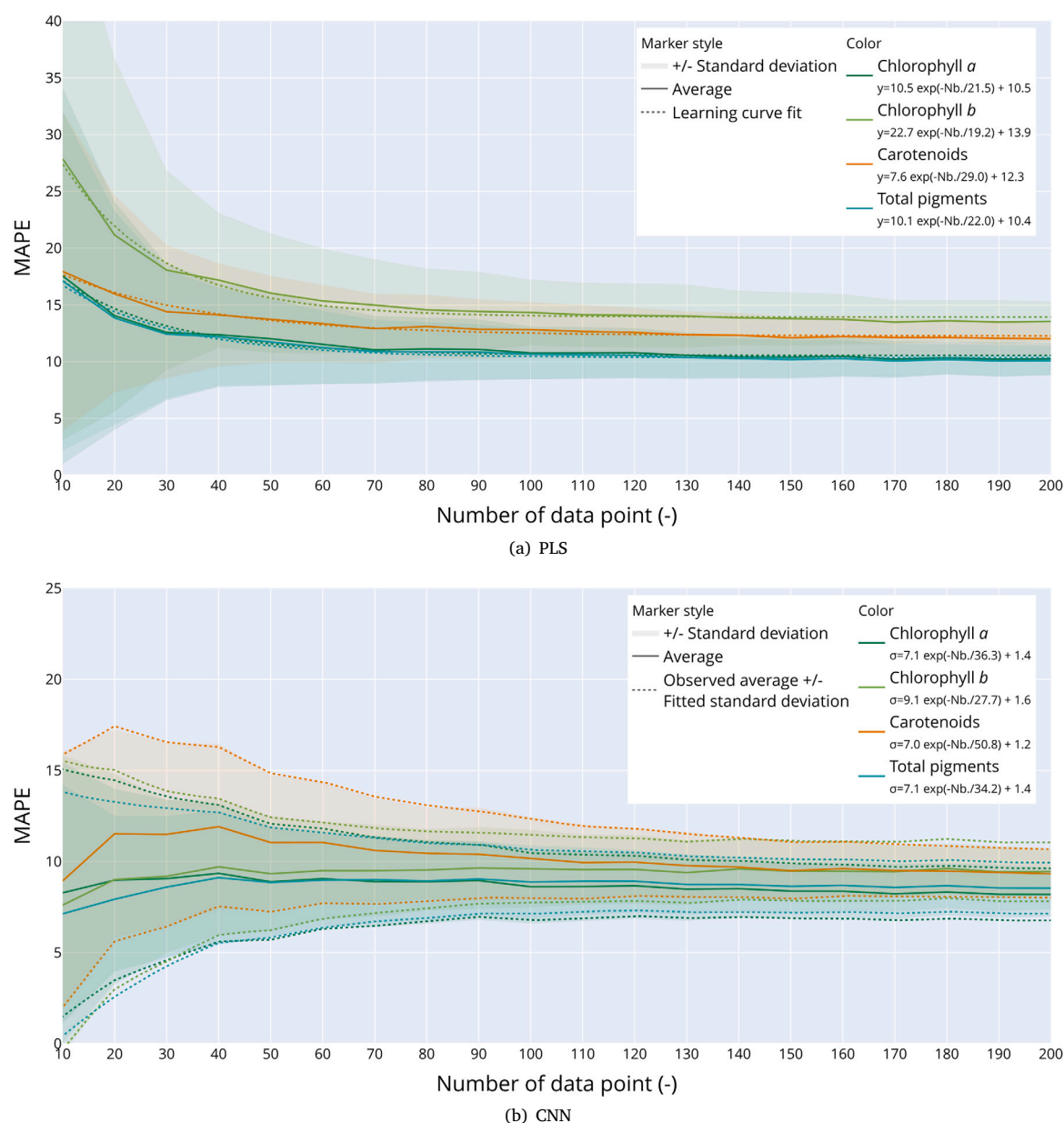
Best performance achieved by the top-performing machine learning pipeline and the top-performing deep learning pipeline. For the CNN data, incremental performance compared to PLS is noted within brackets.

| Algorithm | Dimensions     | MAPE         | RMSE (mg/g)  | NSE (-)       | CCC (-)       |
|-----------|----------------|--------------|--------------|---------------|---------------|
| PLS       | Chlorophyll a  | 10.69        | 2.05         | 0.841         | 0.923         |
|           | Chlorophyll b  | 15.26        | 0.84         | 0.822         | 0.913         |
|           | Carotenoids    | 13.13        | 0.76         | 0.564         | 0.703         |
|           | Total pigments | 10.36        | 3.54         | 0.801         | 0.901         |
| CNN       | Chlorophyll a  | 8.14 (-2.55) | 1.97 (-0.08) | 0.854 (0.013) | 0.923 (0)     |
|           | Chlorophyll b  | 9.61 (-5.65) | 0.66 (-0.18) | 0.888 (0.066) | 0.942 (0.029) |
|           | Carotenoids    | 8.74 (-4.39) | 0.65 (-0.11) | 0.681 (0.117) | 0.789 (0.086) |
|           | Total pigments | 7.69 (-2.67) | 3.21 (-0.33) | 0.836 (0.035) | 0.909 (0.008) |

final performance, depending on the desired outcome, total carotenoids being the most demanding ones.

Moving the CNN learning curves (Fig. 6b), one can see that the convergence dynamic is different from the PLS one. The curves appear

to be relatively flat, hence showing that increasing the number of samples does not improve the average prediction. While counterintuitive at first, it may be explained by the fact that the CNN aggressively targets stable regions of the feature vector (first-order derivative) that contain



**Fig. 6.** Learning curves obtained for the best performing algorithms. (a) - PLS. Fitting procedure performed on the average performance. (b) - CNN. Fitting procedure performed on the standard deviation.

the most informative features, conserved across samples. Yet, it does not mean that the algorithm does not benefit from a higher number of samples. Indeed, while in the PLS case, the standard deviation of the average prediction decreased with the average itself (converging at the same rate), the two indicators show distinct behavior in the case of the CNN. Therefore, the stabilization of the standard deviation can be used as a convergence indicator. Using the 3 times the decay rate approximation to reach 95% of the final performance, proper CNN training can be deemed to require between 80 and 150 samples, for chlorophyll a and chlorophyll b, as lower and upper bounds. Similarly to the metrics obtained on the validation set, four asymptotic average MAPE fall below the 10, with 8.40, 9.52, 9.46, and 8.70 for chlorophyll a, chlorophyll b, and total carotenoids, respectively.

#### 4.4. Limitations & Perspectives

First of all, as shown by the learning curve, it is not reasonable to believe that the opal technique will ever deliver predictions perfectly

matching the values obtained by a proper pigment extraction, even though an infinite amount of data points is available. Then, both procedures are relatively simple to set up and do not require high-end hardware to run. As an illustration, fitting the PLS model is almost instantaneous, while running a CNN inference requires 2 to 3 min on ancient hardware (Intel(R) Core(TM) i9-9880H CPU @ 2.30 GHz, no GPU was used for this study). Still, if one wants to re-optimize the CNN architecture, the required time is, of course, longer, but manageable (about 8 h with the described hardware). Also, CNN training could be sped up by removing the second-order derivatives from the dataset, after confirming that including this part of the dataset did not yield substantial gains, as suggested by the Integrated Gradient analysis.

Another limitation stems from the nature of the biological samples used for this work. Indeed, in green microalgae, chlorophyll a and b are very often correlated, as was the case in this work. This limits the possible contrast in the data set. In addition, as their spectra overlap over the whole visible spectrum, it complicates their distinction and

surely sets a limit to how accurate a spectrophotometric method can be, even when deep learning is resorted to. Furthermore, the reference spectrophotometric method used in this work imposes a limit on the achievable performance. Indeed, while it can be deemed 5% (maximum 10%) accurate for chlorophylls quantification (Porra, 1990), its ability to quantify total carotenoids is less certain (36 MAPE versus HPLC on a previous study carried out by our team (Pozzobon and Camarena-Bernard, 2022)). Beyond the technical observation, this comment can also explain part of the difficulty faced by both machine learning and deep learning algorithms in predicting cells' total carotenoids content.

Finally, while delivering promising results, the proposed method calls for more extensive validation through application to other green microalgae, other microalgae with different pigment profiles, cyanobacteria, colored microorganisms in general, and possibly plant material. To help such a process, the core algorithms (PLS and CNN) are shared, along with the dataset, in the public repository. Therefore, anybody willing to adapt them to new biological material, new solvent (higher polarity shifts peaks to higher wavelengths (Lichtenthaler and Buschmann, 2001)), new spectrophotometer (spectra depend on the apparatus used to produce them, to some extent) should be able to do it easily.

## 5. Conclusion

This article presented a new extraction-free and non-destructive method to evaluate pigment content in microalgae cells. It relies on the use of a spectrophotometer and an optical diffuser, as introduced decades ago by Shibata et al. Yet, it took the technique further ahead by using a solid and easy-to-fit optical diffuser, allowing for reliable measurements. After having solidified the data acquisition procedure, this article investigated how the obtained spectra (for two strains, cultivated in two different vessels) can be linked to microalgae pigment contents obtained by a methanol-extraction method. The retained algorithms were PLS (machine learning) and CNN (deep learning). Overall, augmentation/derivative-based feature engineering is advisable, while signal scaling is only beneficial to PLS. While PLS shows merit, it can sometimes offer some misprediction, resulting in a 10 to 15 MAPE. On the contrary, the CNN achieves a stable sub-10 MAPE, making it the tool of choice to link opal glass spectra and cell pigment content. In the last stage of the study, a learning curve analysis was carried out to demonstrate that the number of samples required to train the algorithms remains manageable (60 to 90 for the PLS, 80 to 150 for the CNN). Overall, this technique reduces pigment quantification from several hours (at best) to less than two minutes and opens the way to reliable online microalgal pigment content monitoring.

## CRedit authorship contribution statement

**Victor Pozzobon:** Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Clarisse Arnoudts:** Writing – review & editing, Investigation. **Wendie Levasseur:** Writing – review & editing, Supervision, Investigation, Funding acquisition.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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## Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.biortech.2026.135966>.

## Data availability

A Python implementation CNN used in this article and the associated data are freely available at <https://github.com/victorpozzobon/opalGlassCNN>.

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